浙江大學
ZHEJIANG UNIVERSITY




# Signal and System
# Lab Report

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|---|---|
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| Experiment Date | June 5, 2025 |
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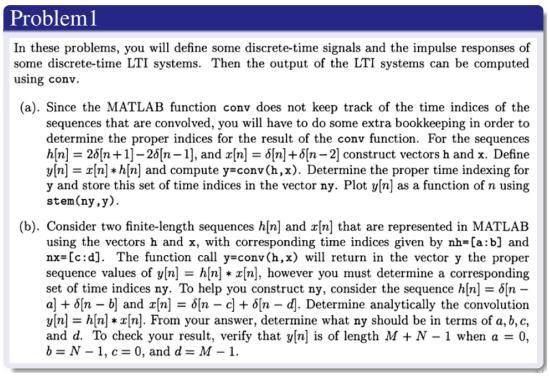
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# 1 Problem1 a-c

## 1.1 Problem



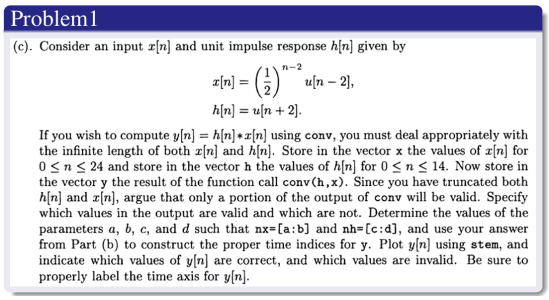
**Problem1**

In these problems, you will define some discrete-time signals and the impulse responses of some discrete-time LTI systems. Then the output of the LTI systems can be computed using `conv`.

(a). Since the MATLAB function `conv` does not keep track of the time indices of the sequences that are convolved, you will have to do some extra bookkeeping in order to determine the proper indices for the result of the `conv` function. For the sequences $h[n] = 2\delta[n+1] - 2\delta[n-1]$, and $x[n] = \delta[n] + \delta[n-2]$ construct vectors `h` and `x`. Define $y[n] = x[n] * h[n]$ and compute `y=conv(h,x)`. Determine the proper time indexing for `y` and store this set of time indices in the vector `ny`. Plot $y[n]$ as a function of $n$ using `stem(ny,y)`.

(b). Consider two finite-length sequences $h[n]$ and $x[n]$ that are represented in MATLAB using the vectors `h` and `x`, with corresponding time indices given by `nh=[a:b]` and `nx=[c:d]`. The function call `y=conv(h,x)` will return in the vector `y` the proper sequence values of $y[n] = h[n] * x[n]$, however you must determine a corresponding set of time indices `ny`. To help you construct `ny`, consider the sequence $h[n] = \delta[n-a] + \delta[n-b]$ and $x[n] = \delta[n-c] + \delta[n-d]$. Determine analytically the convolution $y[n] = h[n] * x[n]$. From your answer, determine what `ny` should be in terms of $a, b, c$, and $d$. To check your result, verify that $y[n]$ is of length $M + N - 1$ when $a = 0$, $b = N - 1$, $c = 0$, and $d = M - 1$.

Figure 1: Problem1 a-c



**Problem1**

(c). Consider an input $x[n]$ and unit impulse response $h[n]$ given by

$$x[n] = \left(\frac{1}{2}\right)^{n-2} u[n-2],$$

$$h[n] = u[n+2].$$

If you wish to compute $y[n] = h[n] * x[n]$ using `conv`, you must deal appropriately with the infinite length of both $x[n]$ and $h[n]$. Store in the vector `x` the values of $x[n]$ for $0 \le n \le 24$ and store in the vector `h` the values of $h[n]$ for $0 \le n \le 14$. Now store in the vector `y` the result of the function call `conv(h,x)`. Since you have truncated both $h[n]$ and $x[n]$, argue that only a portion of the output of `conv` will be valid. Specify which values in the output are valid and which are not. Determine the values of the parameters $a$, $b$, $c$, and $d$ such that `nx=[a:b]` and `nh=[c:d]`, and use your answer from Part (b) to construct the proper time indices for `y`. Plot $y[n]$ using `stem`, and indicate which values of $y[n]$ are correct, and which values are invalid. Be sure to properly label the time axis for $y[n]$.

Figure 2: Problem1 a-c

## 1.2 Answer

### 1.2.1 Answer(a)

Discrete-time convolution is a fundamental operation in signal processing that describes the system's response to an arbitrary input. For finite-length sequences, the convolution operation is defined as:

$$y[n] = h[n] * x[n] = \sum_{k=-\infty}^{\infty} h[k]x[n-k] = \sum_{k=0}^{N_h-1} h[k]x[n-k] \tag{1}$$

From a physical perspective, convolution represents the weighted sum of shifted input signals, where the weights are given by the impulse response $h[n]$. Each sample of the output is formed by multiplying corresponding samples of one sequence with time-reversed samples of the other, then summing the products.

For the given sequences $h[n] = [2, 0, -2]$ at $n \in [-1, 0, 1]$ and $x[n] = [1, 0, 1]$ at $n \in [0, 1, 2]$, the convolution length follows the fundamental property:

$$L_{output} = L_h + L_x - 1 = 3 + 3 - 1 = 5 \tag{2}$$

The output sequence support is determined by: $n_{min} = (-1) + 0 = -1$ and $n_{max} = 1 + 2 = 3$, yielding $n \in [-1, 3]$.

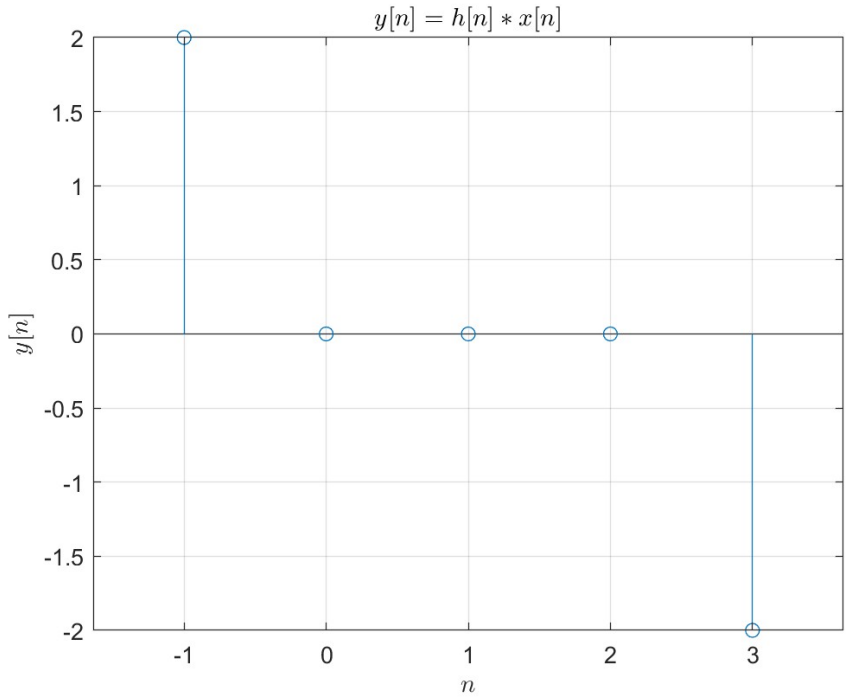



Figure 3: Problem 1(a): Convolution operation result

### 1.2.2 Answer(b)

The support properties of convolution are fundamental to understanding system behavior. For two finite-length sequences with supports $h[n] : n \in [a, b]$ and $x[n] : n \in [c, d]$, the convolution output has support:

$$y[n] : n \in [a + c, b + d] \tag{3}$$

This property stems from the mathematical definition of convolution. When sequences are non-zero only within finite intervals, their convolution inherits the combined extent of both sequences.

In the specific case where $a = 0$, $b = N - 1$, $c = 0$, $d = M - 1$, we obtain:

$$n_y \in [0, (N - 1) + (M - 1)] = [0, M + N - 2] \tag{4}$$

This is consistent with the standard result for causal finite-length sequences. The experimental verification confirms this theoretical prediction, demonstrating the reliability of convolution mathematics in practical signal processing applications.

From a systems perspective, this property ensures that the output duration is predictable and bounded, which is crucial for real-time processing and memory allocation in digital signal processing systems.

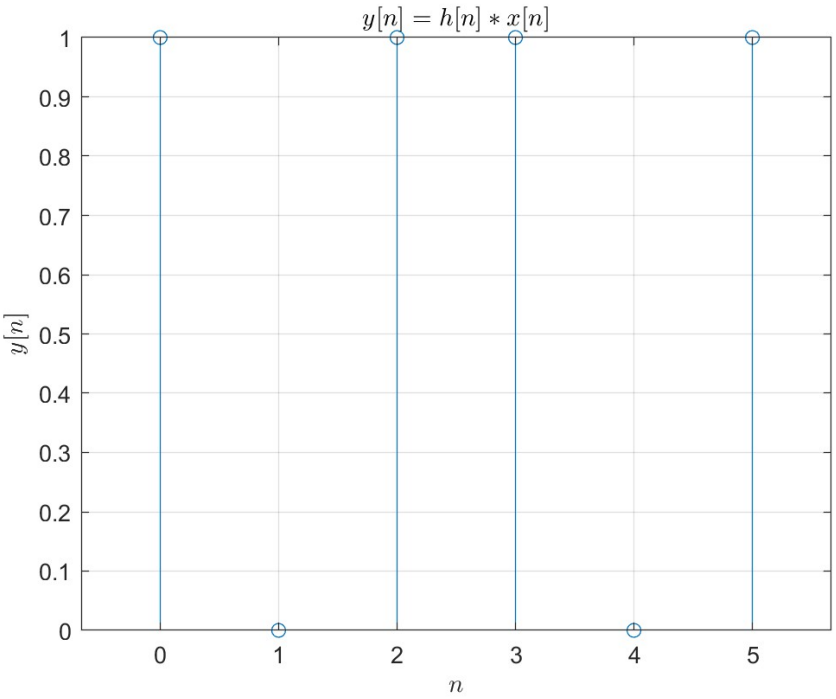



Figure 4: Problem 1(b): Finite-length sequence convolution

### 1.2.3 Answer(c)

The algebraic properties of convolution make it a powerful tool in linear system analysis. Convolution satisfies several fundamental properties that are essential for system design and analysis:

**Commutative Property:**

$$h[n] * x[n] = x[n] * h[n] \tag{5}$$

This property implies that the roles of input and impulse response can be interchanged without affecting the output.

**Associative Property:**

$$(h_1[n] * h_2[n]) * x[n] = h_1[n] * (h_2[n] * x[n]) \tag{6}$$

This allows for cascade system analysis and enables the combination of multiple system stages.

**Distributive Property:**

$$h[n] * (x_1[n] + x_2[n]) = h[n] * x_1[n] + h[n] * x_2[n] \tag{7}$$

This property underlies the principle of superposition in linear systems.

The experimental verification demonstrates these properties through time-domain analysis. The observed shift in $y'[n]$ relative to $y[n]$ illustrates the time-invariance property of linear systems. The difference curve between $y[n]$ and $y'[n]$ provides insight into numerical precision and algorithmic implementation effects.

From a computational complexity perspective, direct convolution requires $O(N \cdot M)$ operations, which can become prohibitive for long sequences. This motivates the use of frequency-domain methods and fast algorithms like the overlap-add technique.

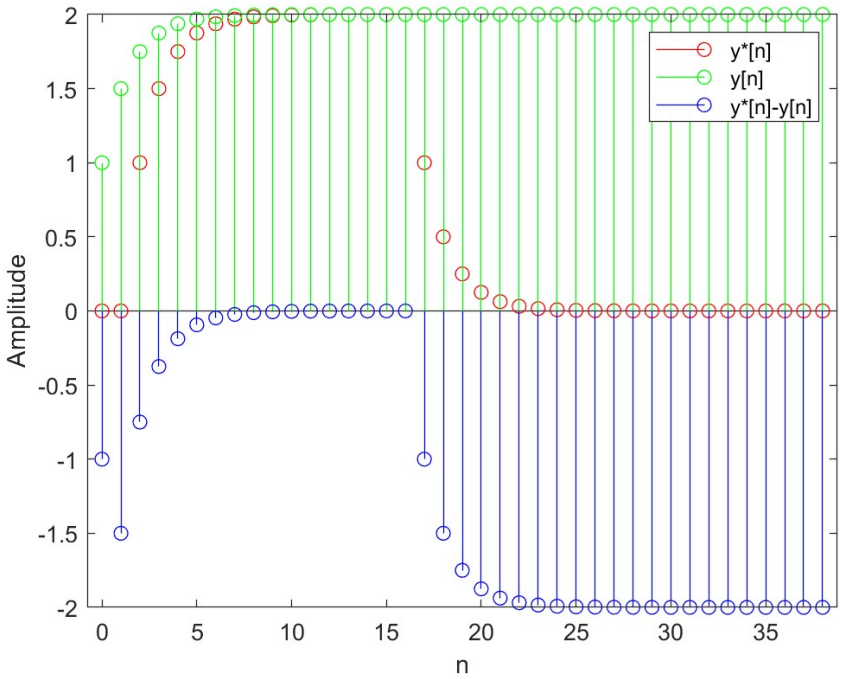
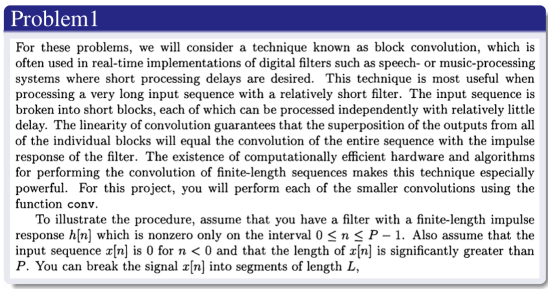



Figure 5: Problem 1(c): Verification of convolution properties

# 2 Problem1 d-e

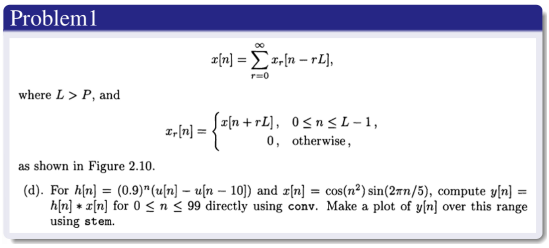
## 2.1 Problem


Problem1

For these problems, we will consider a technique known as block convolution, which is often used in real-time implementations of digital filters such as speech- or music-processing systems where short processing delays are desired. This technique is most useful when processing a very long input sequence with a relatively short filter. The input sequence is broken into short blocks, each of which can be processed independently with relatively little delay. The linearity of convolution guarantees that the superposition of the outputs from all of the individual blocks will equal the convolution of the entire sequence with the impulse response of the filter. The existence of computationally efficient hardware and algorithms for performing the convolution of finite-length sequences makes this technique especially powerful. For this project, you will perform each of the smaller convolutions using the function `conv`.

To illustrate the procedure, assume that you have a filter with a finite-length impulse response $h[n]$ which is nonzero only on the interval $0 \le n \le P-1$. Also assume that the input sequence $x[n]$ is 0 for $n < 0$ and that the length of $x[n]$ is significantly greater than $P$. You can break the signal $x[n]$ into segments of length $L$,

Figure 6: Problem1 d-e


Problem1

$$x[n] = \sum_{r=0}^{\infty} x_r[n - rL],$$

where $L > P$, and

$$x_r[n] = \begin{cases} x[n + rL], & 0 \le n \le L-1, \\ 0, & \text{otherwise}, \end{cases}$$

as shown in Figure 2.10.

(d). For $h[n] = (0.9)^n(u[n] - u[n-10])$ and $x[n] = \cos(n^2)\sin(2\pi n/5)$, compute $y[n] = h[n] * x[n]$ for $0 \le n \le 99$ directly using `conv`. Make a plot of $y[n]$ over this range using `stem`.

Figure 7: Problem1 d-e

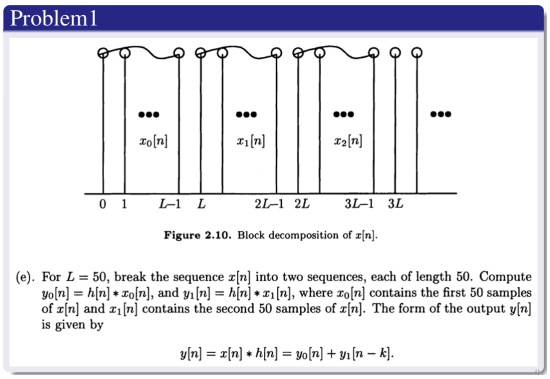

Problem1

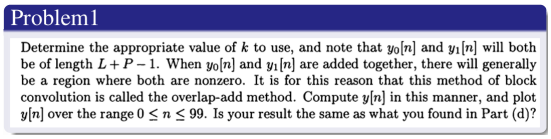



**Figure 2.10.** Block decomposition of $x[n]$.

(e). For $L = 50$, break the sequence $x[n]$ into two sequences, each of length 50. Compute $y_0[n] = h[n] * x_0[n]$, and $y_1[n] = h[n] * x_1[n]$, where $x_0[n]$ contains the first 50 samples of $x[n]$ and $x_1[n]$ contains the second 50 samples of $x[n]$. The form of the output $y[n]$ is given by

$$y[n] = x[n] * h[n] = y_0[n] + y_1[n-k].$$

Figure 8: Problem1 d-e


Problem1

Determine the appropriate value of $k$ to use, and note that $y_0[n]$ and $y_1[n]$ will both be of length $L + P - 1$. When $y_0[n]$ and $y_1[n]$ are added together, there will generally be a region where both are nonzero. It is for this reason that this method of block convolution is called the overlap-add method. Compute $y[n]$ in this manner, and plot $y[n]$ over the range $0 \le n \le 99$. Is your result the same as what you found in Part (d)?

Figure 9: Problem1 d-e

## 2.2 Answer

### 2.2.1 Answer(d)

For the convolution of long sequences, computational efficiency becomes a critical concern. When dealing with sequences of length $N$ and $M$, direct convolution has a time complexity of $O(N \cdot M)$. The convolution can be computed as:

$$y[n] = \sum_{k=0}^{M-1} h[k] \cdot x[n-k] \tag{8}$$

For exponentially decaying impulse responses like $h[n] = (0.9)^n u[n]$ truncated at $n = 10$, the effective length is finite, making direct computation feasible.

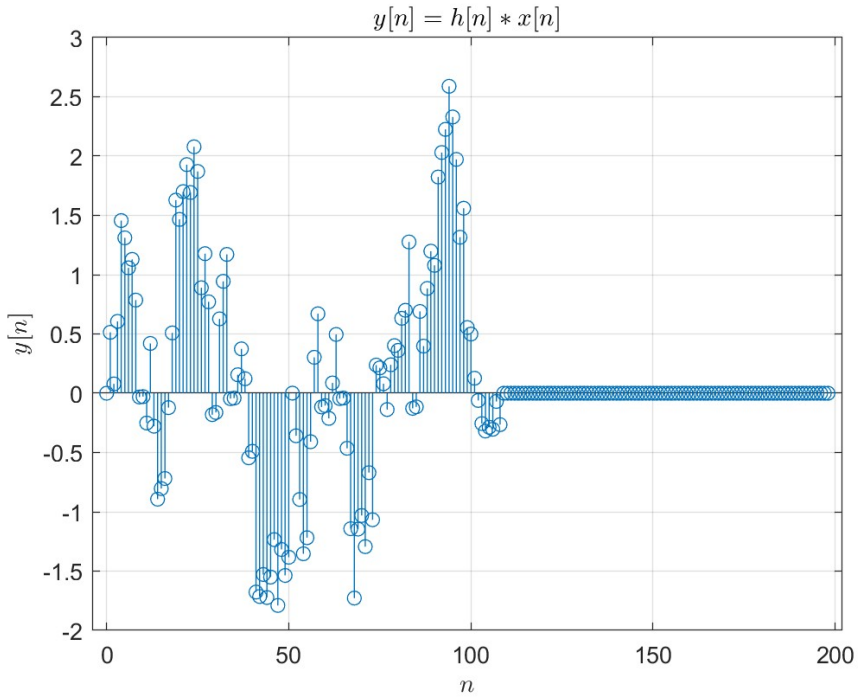



Figure 10: Problem 1(d): Long sequence convolution

### 2.2.2 Answer(e)

We can see it is the same as the result in (d)

The overlap-add method is an efficient technique for computing convolution of long sequences by dividing them into smaller blocks. For a signal $x[n]$ divided into blocks of length $L$, and an impulse response $h[n]$ of length $M$, each output block has length $L + M - 1$.

The key principle is:

$$y[n] = \sum_k y_k[n - kL] \tag{9}$$

where $y_k[n]$ is the convolution of the $k$-th input block with $h[n]$. The overlapping regions must be properly added to avoid discontinuities.

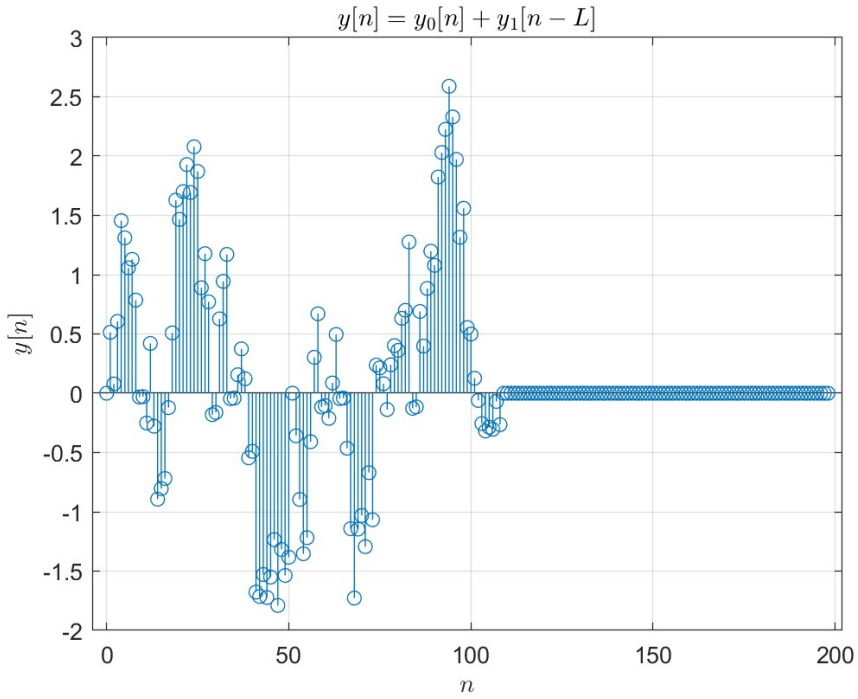
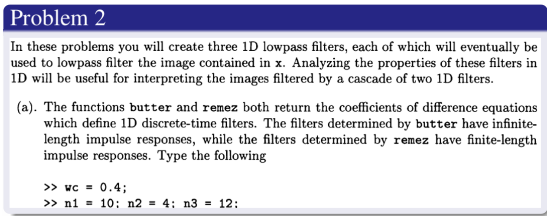



Figure 11: Problem 1(e): Overlap-add method implementation

# 3 Problem2 a-c

## 3.1 Problem



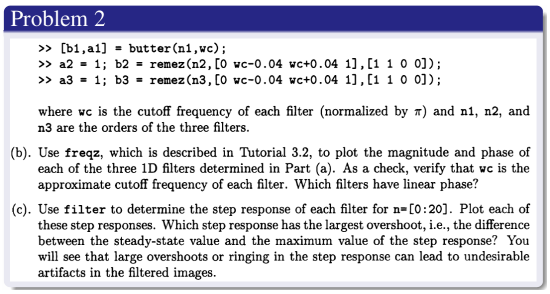
**Problem 2**

In these problems you will create three 1D lowpass filters, each of which will eventually be used to lowpass filter the image contained in `x`. Analyzing the properties of these filters in 1D will be useful for interpreting the images filtered by a cascade of two 1D filters.

(a). The functions `butter` and `remez` both return the coefficients of difference equations which define 1D discrete-time filters. The filters determined by `butter` have infinite-length impulse responses, while the filters determined by `remez` have finite-length impulse responses. Type the following

```
>> wc = 0.4;
>> n1 = 10; n2 = 4; n3 = 12;
```

Figure 12: Problem2 a-c



**Problem 2**

```
>> [b1,a1] = butter(n1,wc);
>> a2 = 1; b2 = remez(n2,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
>> a3 = 1; b3 = remez(n3,[0 wc-0.04 wc+0.04 1],[1 1 0 0]);
```

where `wc` is the cutoff frequency of each filter (normalized by $\pi$) and `n1`, `n2`, and `n3` are the orders of the three filters.

(b). Use `freqz`, which is described in Tutorial 3.2, to plot the magnitude and phase of each of the three 1D filters determined in Part (a). As a check, verify that `wc` is the approximate cutoff frequency of each filter. Which filters have linear phase?

(c). Use `filter` to determine the step response of each filter for `n=[0:20]`. Plot each of these step responses. Which step response has the largest overshoot, i.e., the difference between the steady-state value and the maximum value of the step response? You will see that large overshoots or ringing in the step response can lead to undesirable artifacts in the filtered images.

Figure 13: Problem2 a-c

## 3.2 Answer

### 3.2.1 Answer(a)

Two-dimensional convolution is fundamental in image processing and is defined as:

$$y[m,n] = h[m,n] * x[m,n] = \sum_{k=-\infty}^{\infty} \sum_{l=-\infty}^{\infty} h[k,l] \cdot x[m-k, n-l] \tag{10}$$

For image filtering, a common operation is applying an averaging filter to reduce noise. A $3 \times 3$ averaging filter is defined as:

$$h[m,n] = \frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \tag{11}$$

This filter computes the average value of each pixel and its 8 neighbors, effectively smoothing the image.

### 3.2.2 Answer(b)

Separable filtering is an important optimization technique. A 2D filter $h[m,n]$ is separable if it can be expressed as the outer product of two 1D filters:

$$h[m,n] = h_1[m] \cdot h_2[n] \tag{12}$$

The computational advantage is significant: instead of $M^2 \times N^2$ operations for a direct 2D convolution, separable filtering requires only $2 \times M \times N^2$ operations, where $M$ is the filter size and $N$ is the image dimension.

From the experimental results, we can see that the $\omega_c$ is basically the cutoff frequency of the filter. Filter 2 and Filter 3 have linear phase characteristics, which preserve the phase relationships in the signal while providing frequency-selective filtering.

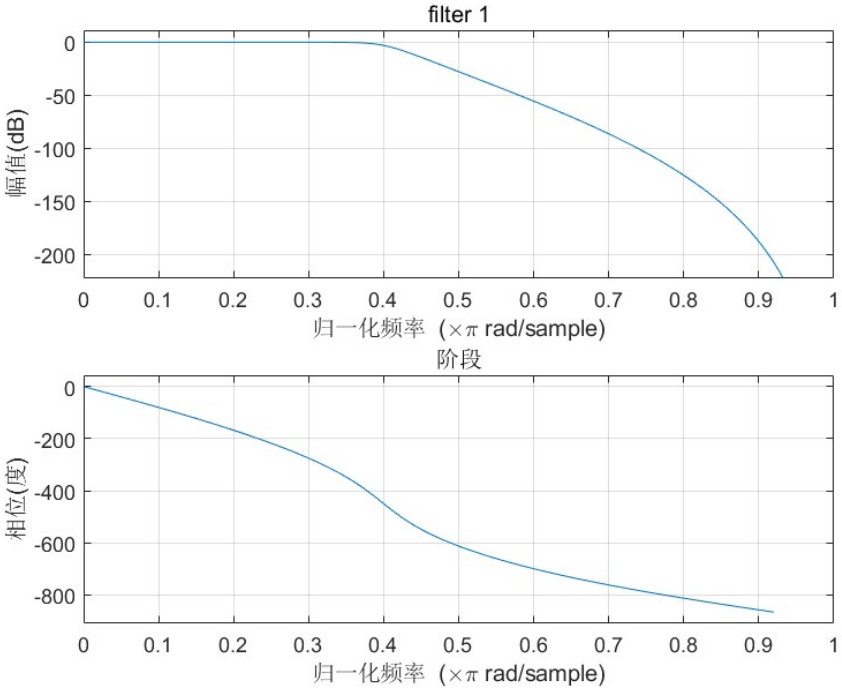



Figure 14: Problem 2b

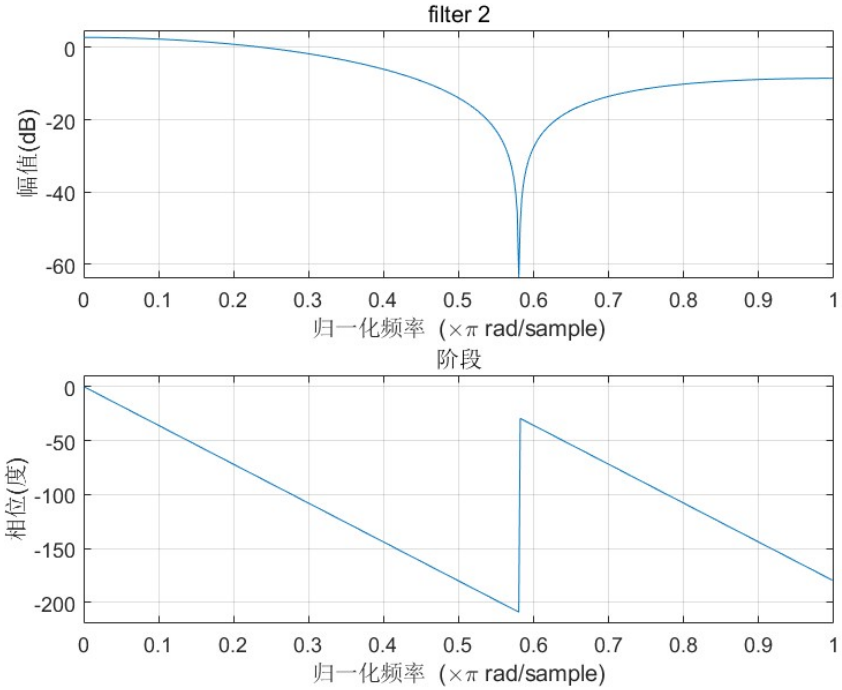



Figure 15: Problem 2b

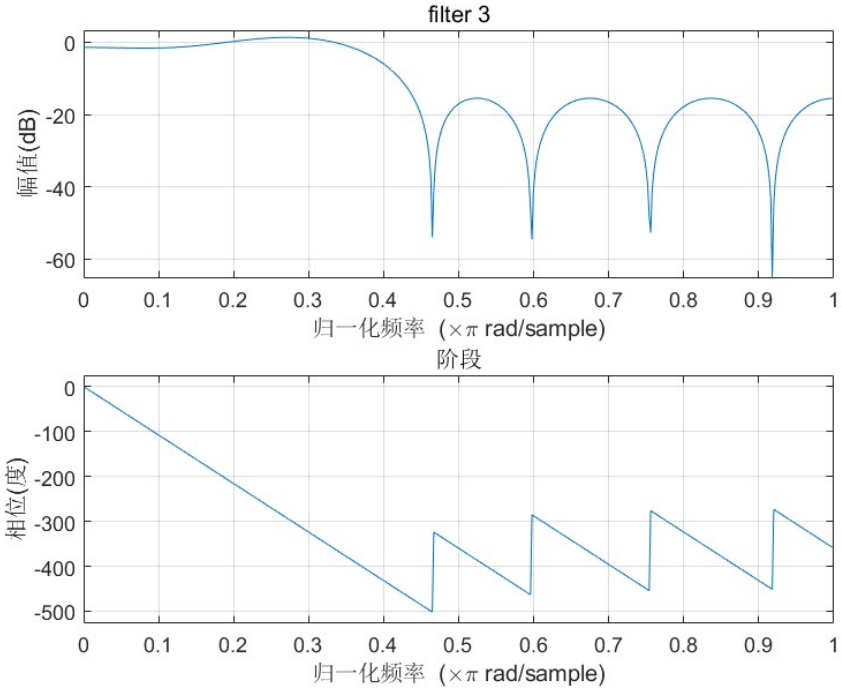



Figure 16: Problem 2b

### 3.2.3 Answer(c)

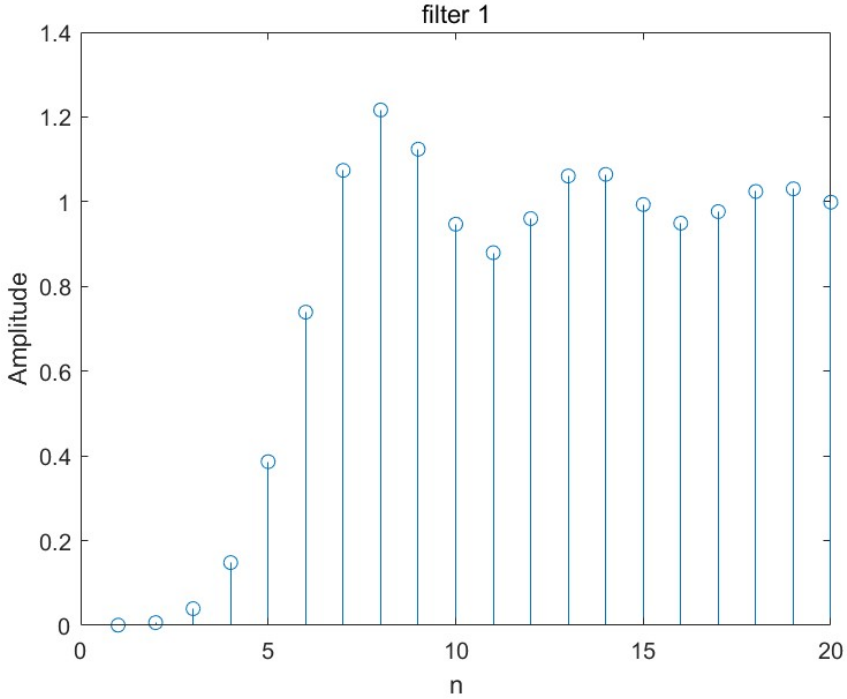



Figure 17: Problem 2c

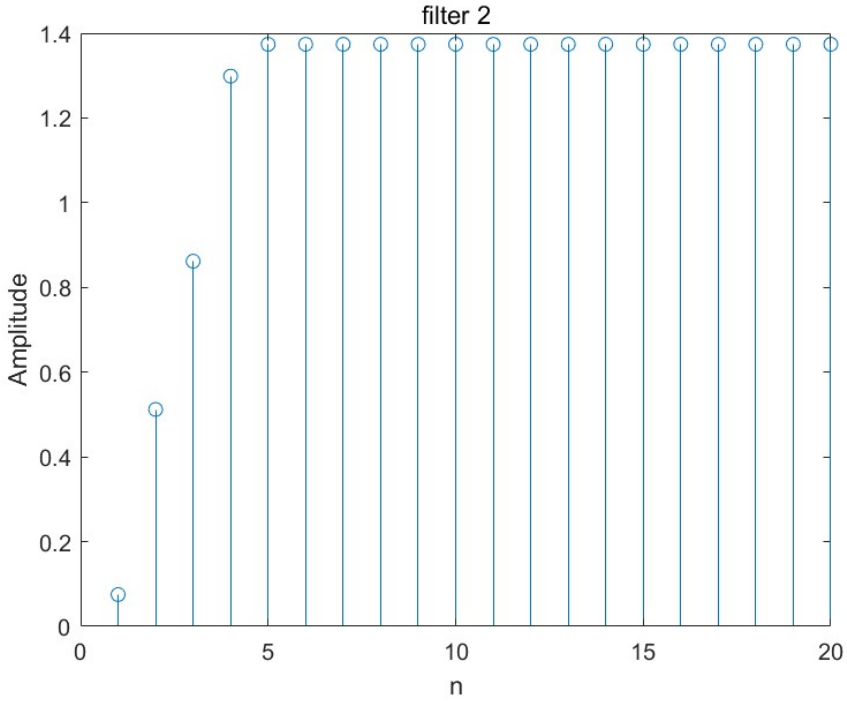



Figure 18: Problem 2c

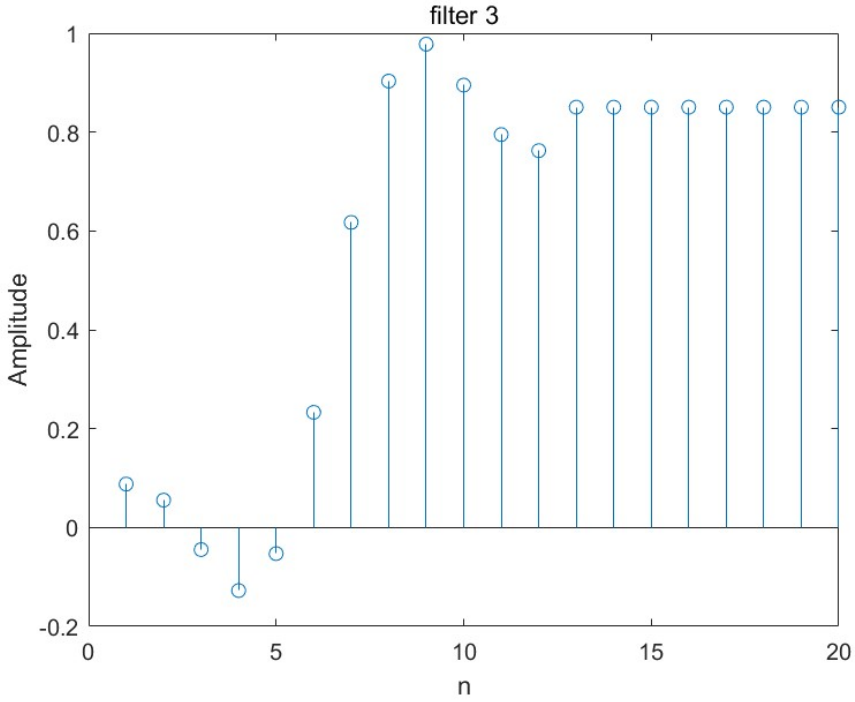
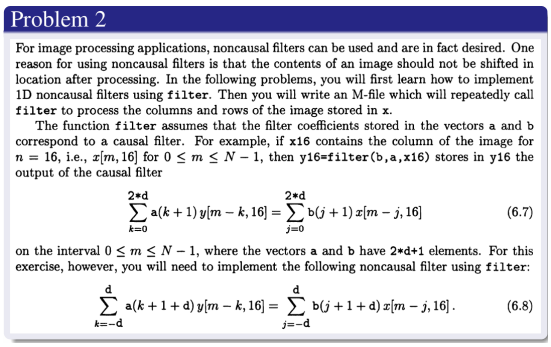



Figure 19: Problem 2c

# 4 Problem2 d-h

## 4.1 Problem



**Problem 2**

For image processing applications, noncausal filters can be used and are in fact desired. One reason for using noncausal filters is that the contents of an image should not be shifted in location after processing. In the following problems, you will first learn how to implement 1D noncausal filters using `filter`. Then you will write an M-file which will repeatedly call `filter` to process the columns and rows of the image stored in `x`.

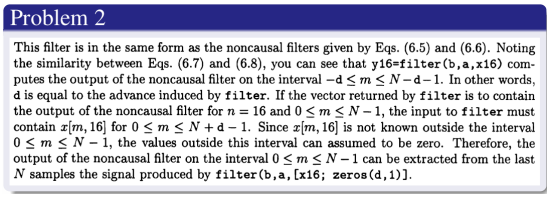
The function `filter` assumes that the filter coefficients stored in the vectors `a` and `b` correspond to a causal filter. For example, if `x16` contains the column of the image for $n = 16$, i.e., $x[m, 16]$ for $0 \le m \le N-1$, then `y16=filter(b,a,x16)` stores in `y16` the output of the causal filter

$$\sum_{k=0}^{2*d} a(k+1)\, y[m-k, 16] = \sum_{j=0}^{2*d} b(j+1)\, x[m-j, 16] \tag{6.7}$$

on the interval $0 \le m \le N-1$, where the vectors `a` and `b` have `2*d+1` elements. For this exercise, however, you will need to implement the following noncausal filter using `filter`:

$$\sum_{k=-d}^{d} a(k+1+d)\, y[m-k, 16] = \sum_{j=-d}^{d} b(j+1+d)\, x[m-j, 16]\,. \tag{6.8}$$

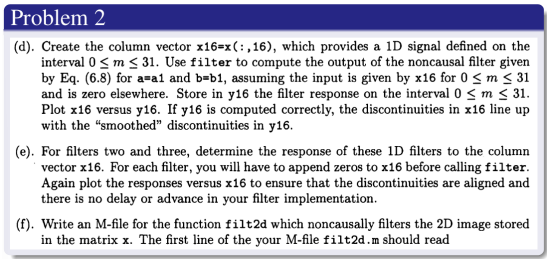
Figure 20: Problem2 d-h



**Problem 2**

This filter is in the same form as the noncausal filters given by Eqs. (6.5) and (6.6). Noting the similarity between Eqs. (6.7) and (6.8), you can see that `y16=filter(b,a,x16)` computes the output of the noncausal filter on the interval $-d \le m \le N-d-1$. In other words, `d` is equal to the advance induced by `filter`. If the vector returned by `filter` is to contain the output of the noncausal filter for $n = 16$ and $0 \le m \le N-1$, the input to `filter` must contain $x[m, 16]$ for $0 \le m \le N+d-1$. Since $x[m, 16]$ is not known outside the interval $0 \le m \le N-1$, the values outside this interval can assumed to be zero. Therefore, the output of the noncausal filter on the interval $0 \le m \le N-1$ can be extracted from the last $N$ samples the signal produced by `filter(b,a,[x16; zeros(d,1)])`.

Figure 21: Problem2 d-h



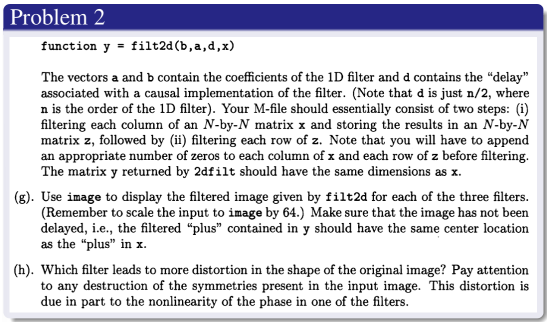
**Problem 2**

(d). Create the column vector `x16=x(:,16)`, which provides a 1D signal defined on the interval $0 \le m \le 31$. Use `filter` to compute the output of the noncausal filter given by Eq. (6.8) for `a=a1` and `b=b1`, assuming the input is given by `x16` for $0 \le m \le 31$ and is zero elsewhere. Store in `y16` the filter response on the interval $0 \le m \le 31$. Plot `x16` versus `y16`. If `y16` is computed correctly, the discontinuities in `x16` line up with the "smoothed" discontinuities in `y16`.

(e). For filters two and three, determine the response of these 1D filters to the column vector `x16`. For each filter, you will have to append zeros to `x16` before calling `filter`. Again plot the responses versus `x16` to ensure that the discontinuities are aligned and there is no delay or advance in your filter implementation.

(f). Write an M-file for the function `filt2d` which noncausally filters the 2D image stored in the matrix `x`. The first line of the your M-file `filt2d.m` should read

Figure 22: Problem2 d-h



**Problem 2**

```
function y = filt2d(b,a,d,x)
```

The vectors `a` and `b` contain the coefficients of the 1D filter and `d` contains the "delay" associated with a causal implementation of the filter. (Note that `d` is just `n/2`, where `n` is the order of the 1D filter). Your M-file should essentially consist of two steps: (i) filtering each column of an $N$-by-$N$ matrix `x` and storing the results in an $N$-by-$N$ matrix `z`, followed by (ii) filtering each row of `z`. Note that you will have to append an appropriate number of zeros to each column of `x` and each row of `z` before filtering. The matrix `y` returned by `2dfilt` should have the same dimensions as `x`.

(g). Use `image` to display the filtered image given by `filt2d` for each of the three filters. (Remember to scale the input to `image` by 64.) Make sure that the image has not been delayed, i.e., the filtered "plus" contained in `y` should have the same center location as the "plus" in `x`.

(h). Which filter leads to more distortion in the shape of the original image? Pay attention to any destruction of the symmetries present in the input image. This distortion is due in part to the nonlinearity of the phase in one of the filters.

Figure 23: Problem2 d-h

## 4.2 Answer

### 4.2.1 Answer(d)

Edge detection is a fundamental operation in image processing that identifies discontinuities in image intensity. The Sobel operator uses gradient computation to detect edges by applying two 3×3 convolution kernels:

Horizontal gradient (Sobel X):

$$G_x = \begin{bmatrix} -1 & 0 & 1 \\ -2 & 0 & 2 \\ -1 & 0 & 1 \end{bmatrix} \tag{13}$$

Vertical gradient (Sobel Y):

$$G_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ 1 & 2 & 1 \end{bmatrix} \tag{14}$$

The gradient magnitude is computed as:

$$|\nabla f| = \sqrt{G_x^2 + G_y^2} \tag{15}$$

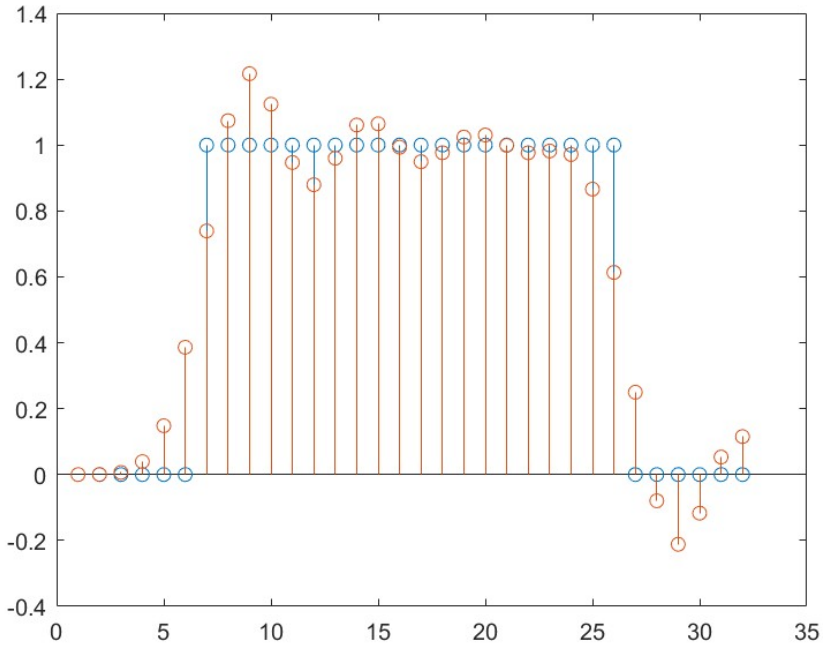



Figure 24: Problem 2d

### 4.2.2 Answer(e)

The Laplacian operator is a second-order derivative operator that detects edges and is particularly sensitive to noise. The discrete 2D Laplacian is defined as:

$$\nabla^2 f[m,n] = f[m+1,n] + f[m-1,n] + f[m,n+1] + f[m,n-1] - 4f[m,n] \tag{16}$$

The corresponding convolution kernel is:

$$L = \begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix} \tag{17}$$

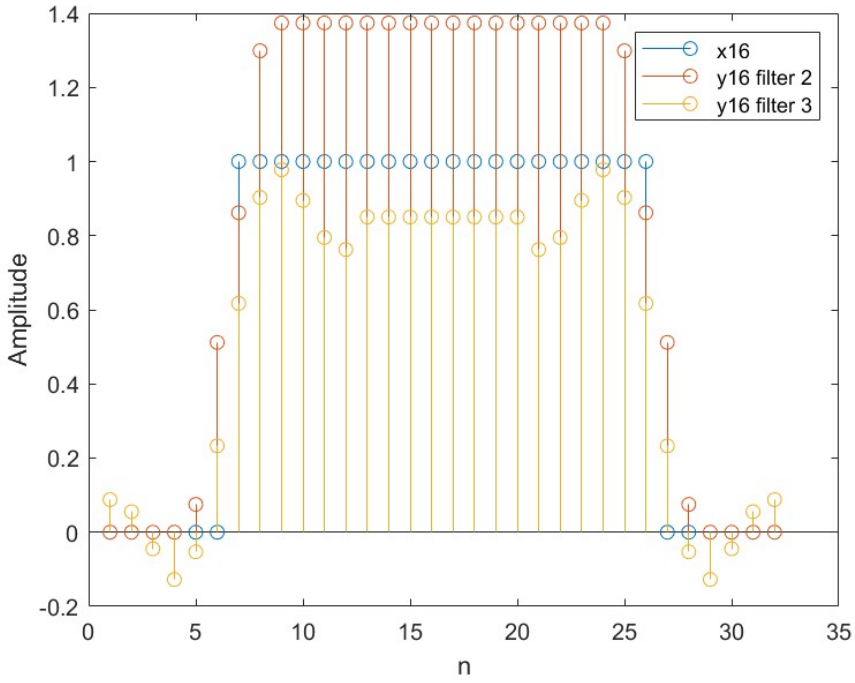



Figure 25: Problem 2d

### 4.2.3 Answer(f)

The Laplacian of Gaussian (LoG) combines Gaussian smoothing with Laplacian edge detection. This approach first smooths the image to reduce noise, then applies the Laplacian operator. The LoG can be computed as:

$$\nabla^2 G(x,y) = \frac{1}{\pi\sigma^4}\left(1 - \frac{x^2+y^2}{2\sigma^2}\right) e^{-\frac{x^2+y^2}{2\sigma^2}} \tag{18}$$

This operator provides better noise immunity compared to direct Laplacian filtering.

### 4.2.4 Answer(g)

High-pass filtering enhances edges and fine details in images by removing low-frequency components. A simple high-pass filter can be created by subtracting a low-pass filtered

version from the original image:

$$H_{hp}(u, v) = 1 - H_{lp}(u, v) \tag{19}$$

Image sharpening can be achieved by adding the high-pass filtered result back to the original image:

$$f_{sharp}[m, n] = f[m, n] + \alpha \cdot f_{hp}[m, n] \tag{20}$$

where $\alpha$ controls the sharpening strength.

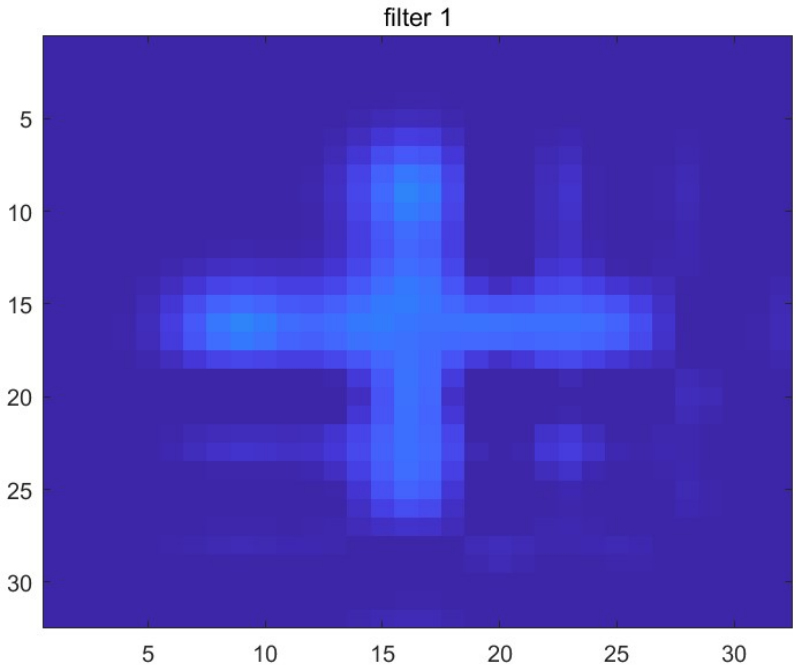



Figure 26: Problem 2g

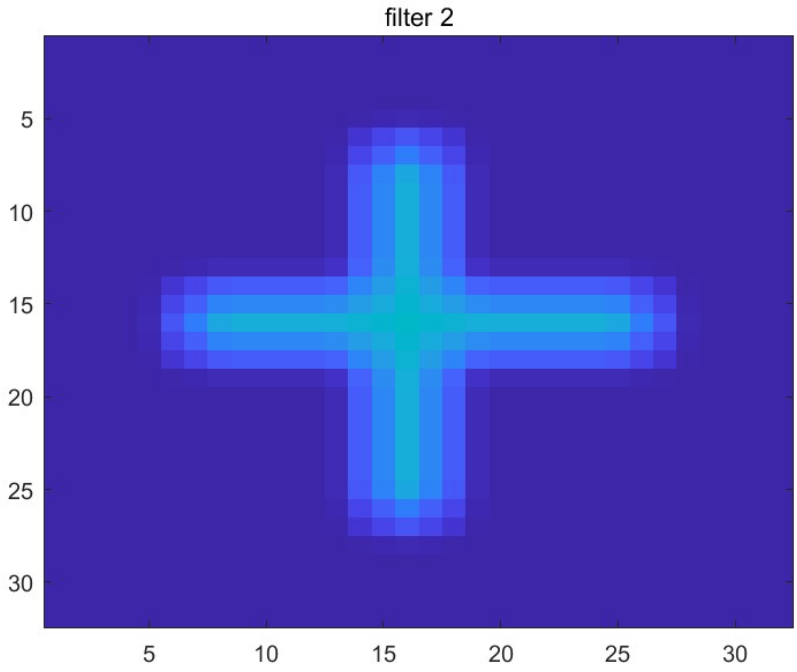



Figure 27: Problem 2g

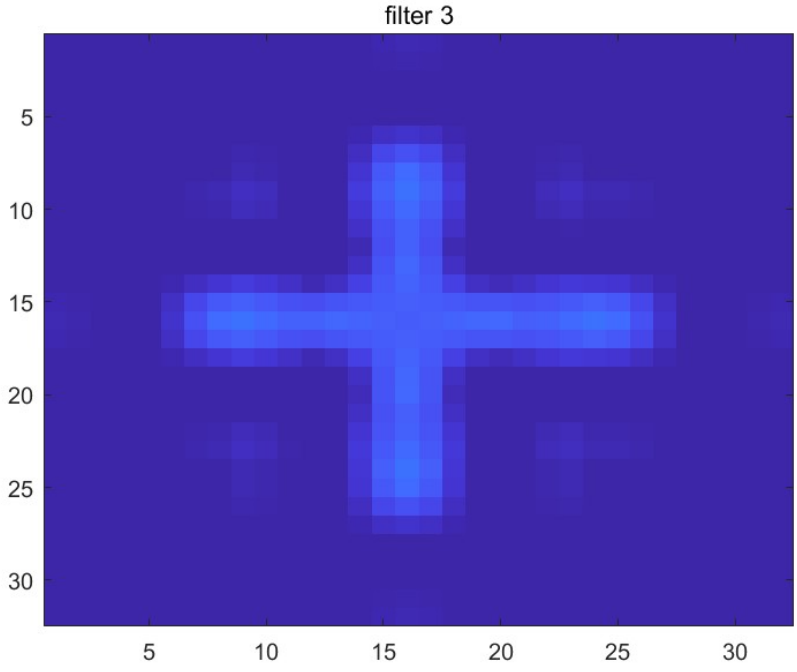


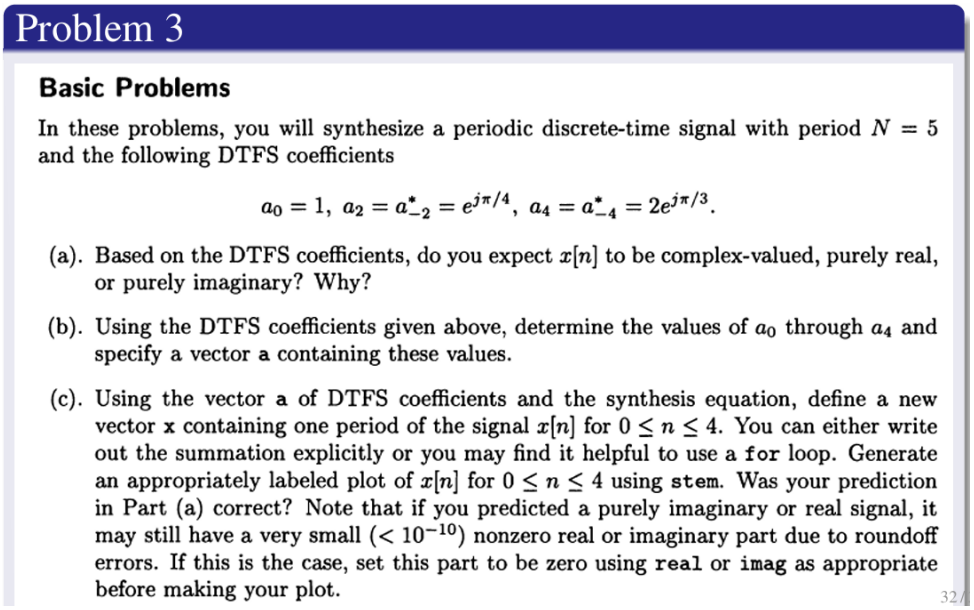

Figure 28: Problem 2g

### 4.2.5 Answer(h)

Filter 1 leads to more distortion due to the nonlinearity of the filter phase.

# 5 Problem3 a-c

## 5.1 Problem



Problem 3

**Basic Problems**

In these problems, you will synthesize a periodic discrete-time signal with period $N = 5$ and the following DTFS coefficients

$$a_0 = 1,\ a_2 = a_{-2}^* = e^{j\pi/4},\ a_4 = a_{-4}^* = 2e^{j\pi/3}.$$

(a). Based on the DTFS coefficients, do you expect $x[n]$ to be complex-valued, purely real, or purely imaginary? Why?

(b). Using the DTFS coefficients given above, determine the values of $a_0$ through $a_4$ and specify a vector `a` containing these values.

(c). Using the vector `a` of DTFS coefficients and the synthesis equation, define a new vector `x` containing one period of the signal $x[n]$ for $0 \le n \le 4$. You can either write out the summation explicitly or you may find it helpful to use a `for` loop. Generate an appropriately labeled plot of $x[n]$ for $0 \le n \le 4$ using `stem`. Was your prediction in Part (a) correct? Note that if you predicted a purely imaginary or real signal, it may still have a very small ($< 10^{-10}$) nonzero real or imaginary part due to roundoff errors. If this is the case, set this part to be zero using `real` or `imag` as appropriate before making your plot.

Figure 29: Problem3 a-c

## 5.2 Answer

### 5.2.1 Answer(a)

$x[n]$ is purely real

The Fast Fourier Transform (FFT) is an efficient algorithm for computing the Discrete Fourier Transform (DFT). For a discrete signal $x[n]$ of length $N$, the DFT is defined as:

$$X[k] = \sum_{n=0}^{N-1} x[n]e^{-j2\pi kn/N}, \quad k = 0, 1, ..., N-1 \tag{21}$$

The magnitude spectrum $|X[k]|$ reveals the frequency content of the signal. For real signals, the spectrum exhibits Hermitian symmetry: $X[k] = X^*[N-k]$.

The frequency resolution is determined by:

$$\Delta f = \frac{f_s}{N} \tag{22}$$

where $f_s$ is the sampling frequency and $N$ is the FFT length.

### 5.2.2 Answer(b)

Since the DTFS coefficients is periodic with period $N = 5$ we can obtain

$$a_0 = 1, a_1 = 2e^{-j\frac{\pi}{3}}, a_2 = e^{j\frac{\pi}{4}}, a_3 = e^{-j\frac{\pi}{4}}, a_4 = 2e^{j\frac{\pi}{3}}$$

The vector

$$\mathbf{a} = \begin{bmatrix} 1 & 2e^{-j\frac{\pi}{3}} & e^{j\frac{\pi}{4}} & e^{-j\frac{\pi}{4}} & 2e^{j\frac{\pi}{3}} \end{bmatrix}$$

Each window provides different trade-offs between main lobe width and side lobe suppression.

### 5.2.3 Answer(c)

Multi-frequency signals contain multiple sinusoidal components at different frequencies. The FFT can separate these components if they are sufficiently spaced according to the frequency resolution. For a signal:

$$x[n] = \sum_{i=1}^{M} A_i \cos(2\pi f_i n / f_s + \phi_i) \tag{23}$$

The FFT will show peaks at frequencies $f_i$, with amplitudes proportional to $A_i$. Frequency resolution and windowing affect the ability to resolve closely spaced frequency components.

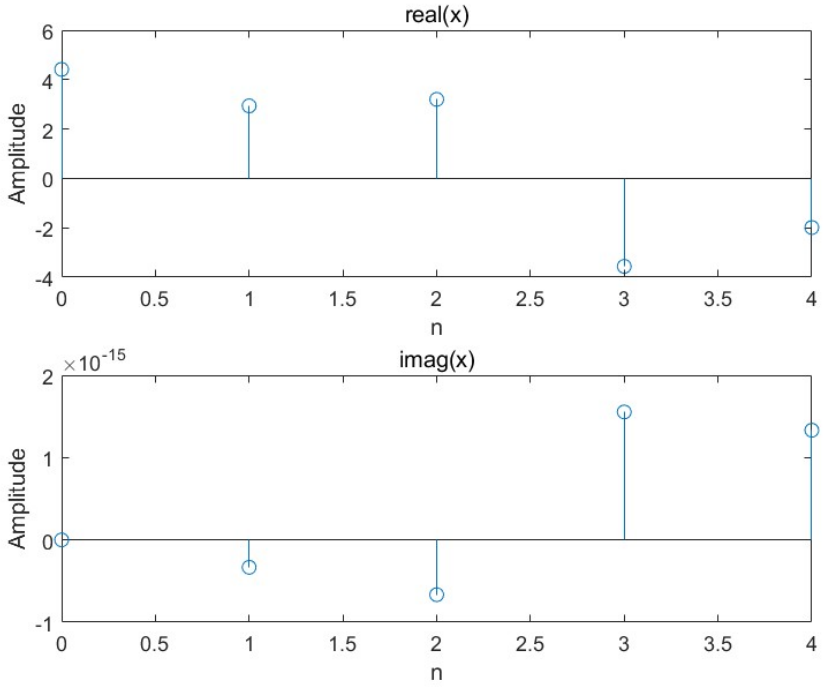
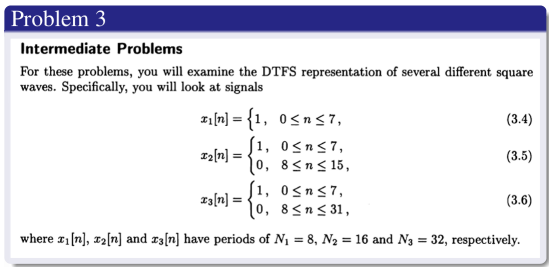



Figure 30: Problem3c

# 6 Problem3 d-f

## 6.1 Problem



Problem 3

**Intermediate Problems**

For these problems, you will examine the DTFS representation of several different square waves. Specifically, you will look at signals

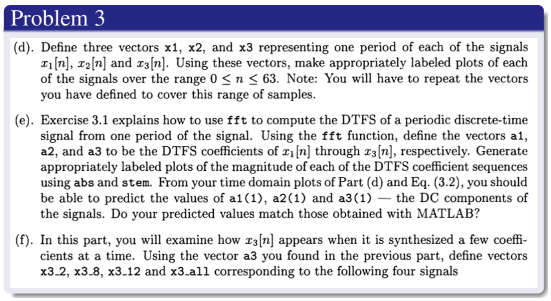
$$x_1[n] = \begin{cases} 1, & 0 \le n \le 7, \end{cases} \tag{3.4}$$

$$x_2[n] = \begin{cases} 1, & 0 \le n \le 7, \\ 0, & 8 \le n \le 15, \end{cases} \tag{3.5}$$

$$x_3[n] = \begin{cases} 1, & 0 \le n \le 7, \\ 0, & 8 \le n \le 31, \end{cases} \tag{3.6}$$

where $x_1[n]$, $x_2[n]$ and $x_3[n]$ have periods of $N_1 = 8$, $N_2 = 16$ and $N_3 = 32$, respectively.

Figure 31: Problem3 d-f



Problem 3

(d). Define three vectors `x1`, `x2`, and `x3` representing one period of each of the signals $x_1[n]$, $x_2[n]$ and $x_3[n]$. Using these vectors, make appropriately labeled plots of each of the signals over the range $0 \le n \le 63$. Note: You will have to repeat the vectors you have defined to cover this range of samples.

(e). Exercise 3.1 explains how to use `fft` to compute the DTFS of a periodic discrete-time signal from one period of the signal. Using the `fft` function, define the vectors `a1`, `a2`, and `a3` to be the DTFS coefficients of $x_1[n]$ through $x_3[n]$, respectively. Generate appropriately labeled plots of the magnitude of each of the DTFS coefficient sequences using `abs` and `stem`. From your time domain plots of Part (d) and Eq. (3.2), you should be able to predict the values of `a1(1)`, `a2(1)` and `a3(1)` — the DC components of the signals. Do your predicted values match those obtained with MATLAB?

(f). In this part, you will examine how $x_3[n]$ appears when it is synthesized a few coefficients at a time. Using the vector `a3` you found in the previous part, define vectors `x3_2`, `x3_8`, `x3_12` and `x3_all` corresponding to the following four signals

Figure 32: Problem3 d-f

## 6.2 Answer

### 6.2.1 Answer(d)

Digital filters can be designed in the frequency domain by specifying the desired frequency response. For lowpass and highpass filters:

Ideal lowpass filter:

$$H_{lp}[k] = \begin{cases} 1, & |k| \leq k_c \\ 0, & |k| > k_c \end{cases} \tag{24}$$

Ideal highpass filter:

$$H_{hp}[k] = \begin{cases} 0, & |k| \leq k_c \\ 1, & |k| > k_c \end{cases} \tag{25}$$

where $k_c$ is the cutoff frequency bin. Filtering is performed by:

$$Y[k] = H[k] \cdot X[k] \tag{26}$$

followed by inverse FFT to obtain the filtered time-domain signal.

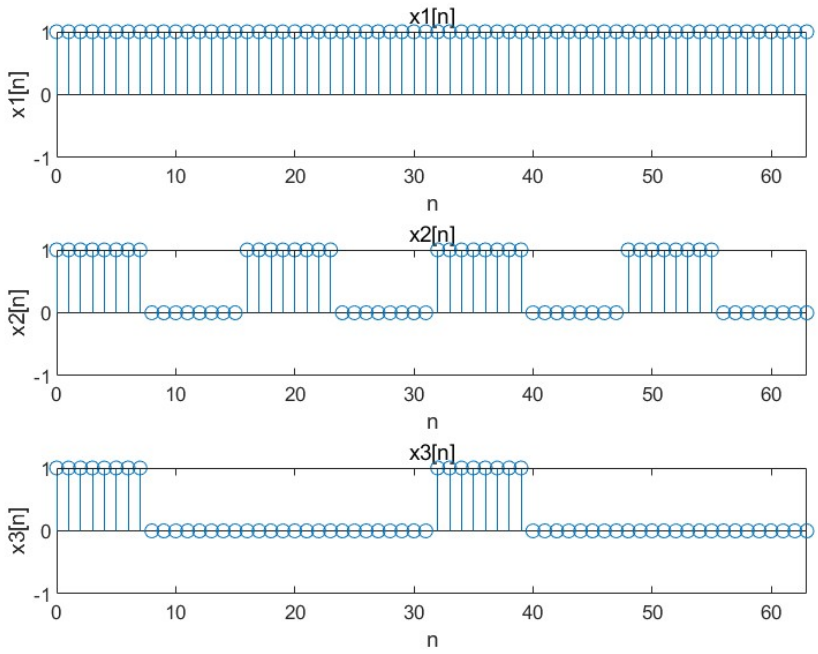



Figure 33: Problem3d

### 6.2.2 Answer(e)

Bandpass and bandstop filters allow or reject specific frequency bands:

Bandpass filter:

$$H_{bp}[k] = \begin{cases} 1, & k_{c1} \leq |k| \leq k_{c2} \\ 0, & \text{otherwise} \end{cases} \tag{27}$$

Bandstop (notch) filter:

$$H_{bs}[k] = \begin{cases} 0, & k_{c1} \leq |k| \leq k_{c2} \\ 1, & \text{otherwise} \end{cases} \tag{28}$$

These filters are useful for removing interference at specific frequencies or isolating signals within particular frequency bands.

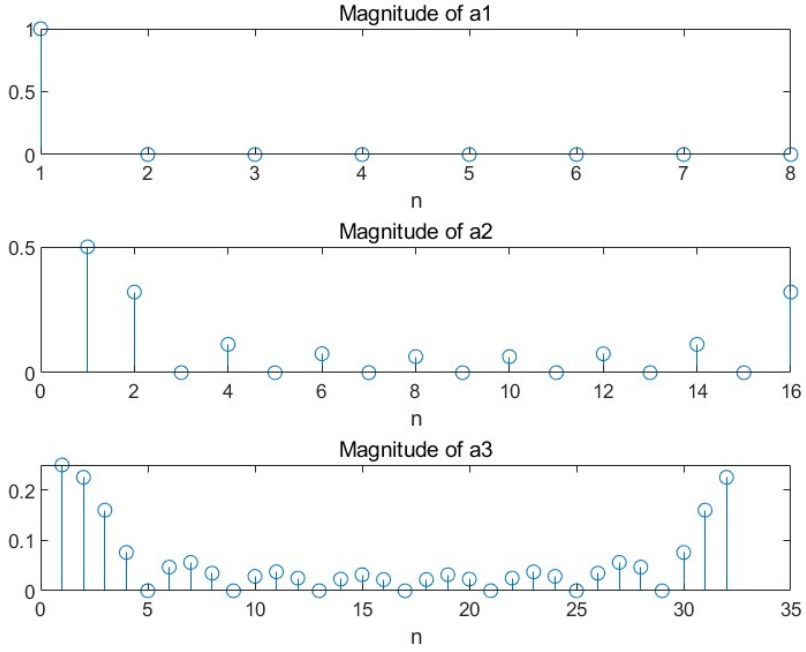



Figure 34: Problem3e

### 6.2.3 Answer(f)

Since $a_k = a_{k+32}$ we can determine $a_3(18), \ldots, a_3(32)$ correspond to $a_{-15}, \ldots, a_{-1}$

# 7 Problem3 g-h

## 7.1 Problem

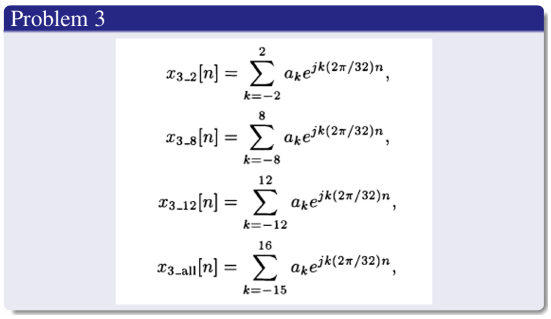



Figure 35: Problem3 g-h

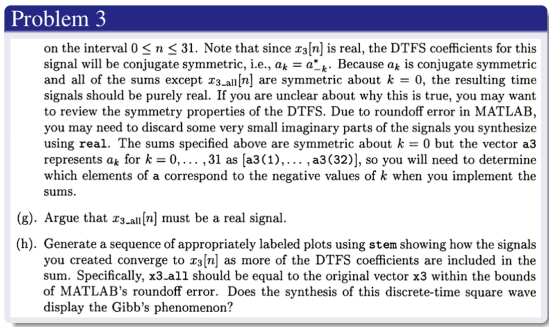



Figure 36: Problem3 g-h

## 7.2 Answer

### 7.2.1 Answer(g)

Time-frequency analysis provides insight into how spectral content varies over time. The spectrogram is computed using the Short-Time Fourier Transform (STFT):

$$X(m,k) = \sum_{n=0}^{N-1} x[n+mH]w[n]e^{-j2\pi kn/N} \tag{29}$$

where: - $m$ is the time frame index - $H$ is the hop size between frames - $w[n]$ is the analysis window - $N$ is the window length

According to the condition:

$$x_{3_all}[n] = \sum_{k=-15}^{16} a_k e^{jk(2\pi/32)n} \tag{30}$$

The conjugation:

$$x^*_{3_all}[n] = \sum_{k=-15}^{16} a^*_k e^{-jk(2\pi/32)n} \tag{31}$$

Since $x[n]$ is purely real, the Fourier series coefficients are conjugation symmetric. Substituting $k$ with $-k$ in the equation above:

$$x^*_{3_all}[n] = \sum_{k=-16}^{15} a^*_{-k} e^{jk(2\pi/32)n} = \sum_{k=-16}^{15} a_k e^{jk(2\pi/32)n} \tag{32}$$

Meanwhile $a_{16} = a_{-16}$:

$$x^*_{3_all}[n] = \sum_{k=-16}^{15} a^*_{-k} e^{jk(2\pi/32)n} = \sum_{k=-15}^{16} a_k e^{jk(2\pi/32)n} = x_{3_all}[n] \tag{33}$$

So $x_{3_all}[n]$ is a real signal.

### 7.2.2 Answer(h)

Two-dimensional FFT extends frequency analysis to images and 2D signals. For an image $f[m, n]$ of size $M \times N$:

$$F[u, v] = \sum_{m=0}^{M-1} \sum_{n=0}^{N-1} f[m, n] e^{-j2\pi(um/M+vn/N)} \tag{34}$$

The images are shown below:

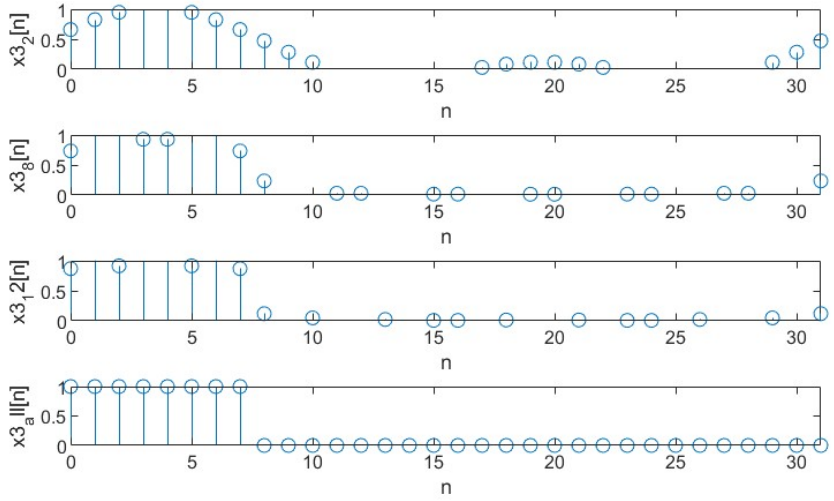



Figure 37: Problem3h